

Posets

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1 Notation

$\mathbb{N}$ = Natural Numbers

$\mathbb{P}$ = Positive Integers

$\mathbb{Z}$ = Integers

$\mathbb{R}$ = Real Numbers

$\mathbb{C}$ = Complex Numbers

$\mathbf{R}$ = Commutative Unital RING

$\mathcal{P}$ = Set of all Prime numbers

$[a, b] = \{x \in \mathbb{Z} \mid a \leq x \leq b\}$

$[a] = [1, a]$

$\xrightarrow{\sim}$ = Used to represent a bijective function (one-one mapping)

$\xrightarrow[\phi]{\sim}$ = Used to show the bijective mapping under the function ϕ

$\hat{0}$ = Smallest element of the Poset

$\hat{1}$ = Greatest element of the Poset

$\leq_X$ = less than or equal, in the poset X .

$n(\dots)$ = cardinality of the expression in the parenths.

2 Partially Ordered Sets / Posets

A partially ordered set also known as a Poset is a pair of a set P and a relation $\leq: P^2$ s.t it satisfies:

1. $\forall x \in P. x \leq x$ (Reflexivity)
2. $\forall x, y, z \in P, x \leq y, y \leq z. x \leq z$. (Transitivity)
3. $\forall x, y \in P, x \leq y, y \leq x. x = y$. (Anti-symmetry)

Common examples of Posets include

1. all subsets of $(\mathbb{C}, \leq)$
2. $(\mathbb{P}, |)$
3. $\forall a, b \in \mathbb{Z}. [a, b]$
4. $\forall X : \text{Set}. (2^X, \subseteq)$
5. $(\mathbb{R}^2, \{(x, y), (x', y') \in \mathbb{R}^2 \mid x \leq x', y \leq y'\})$

Chains and Anti-chains

A Poset $(P, \leq)$ is called a chain if $\forall x, y \in P, x \leq y$ or $y \leq x$. Common examples of this would include $\mathbb{N}, \mathbb{P}, \mathbb{R}$ and their subsets.

A Poset $(P, \leq)$ is called an anti-chain if $\forall x, y \in P, x \leq y \iff y \leq x$. A common example of an antichain would be the set of all human languages, or programming languages.

2.1 Hasse Diagrams

A common way to graphically represent a Poset is by using Hasse diagrams. A Hasse diagram is a graphical representation of a Poset where the following conditions are met:

1. Vertices of the graph represent an element of the poset P
2. There is an edge from x to y if x covers y .

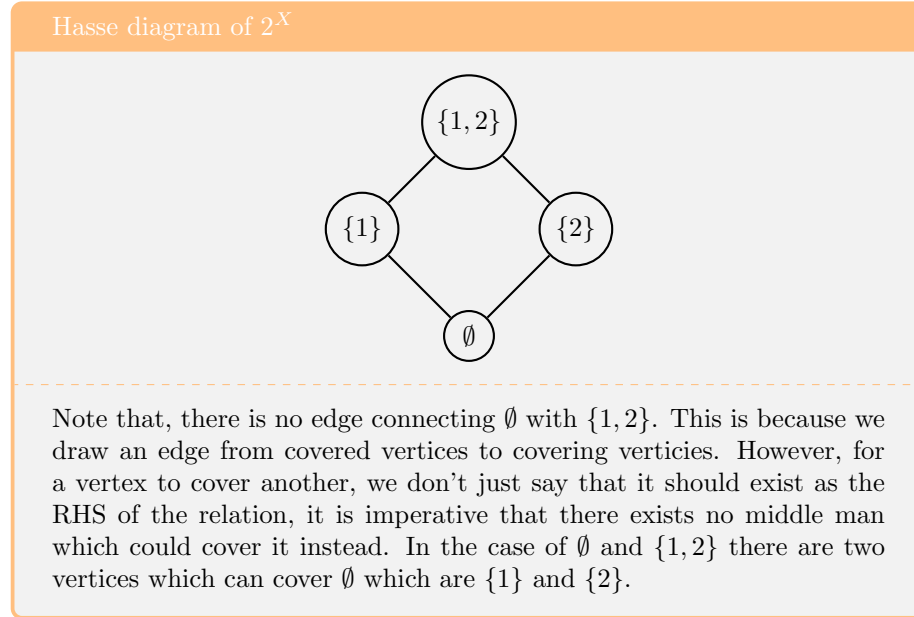
Cover here means the following: If $\forall x, y \in P. \nexists z \in P, z \leq x, y \leq z, x \neq z, y \neq z$. then x is said to cover y .

Generally, Hasse diagrams are drawn without arrows, by placing the covering element physically above the covered element.

For example: Consider the Poset $(2^{[2]}, \subseteq)$ The containment relation can be listed as:

X	$Y \in 2^{[2]} \text{ s.t. } X \subseteq Y$
$\emptyset$	$\emptyset, \{1\}, \{2\}, \{1, 2\}$
$\{1\}$	$\{1\}, \{1, 2\}$
$\{2\}$	$\{2\}, \{1, 2\}$
$\{1, 2\}$	$\{1, 2\}$

This would be drawn in a Hasse diagram as follows:

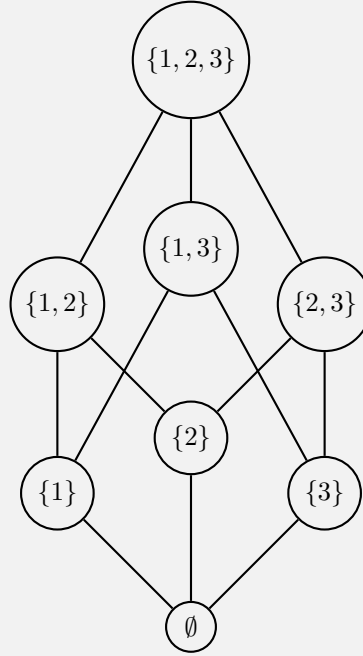


Similarly now let's look at Hasse diagram for $([3], \subseteq)$

We get the table as:

X	$Y \in 2^{[3]} \text{ s.t. } X \subseteq Y$
$\emptyset$	$\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}$
$\{1\}$	$\{1\}, \{1, 2\}, \{1, 3\}, \{1, 2, 3\}$
$\{2\}$	$\{2\}, \{1, 2\}, \{2, 3\}, \{1, 2, 3\}$
$\{3\}$	$\{3\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}$
$\{1, 2\}$	$\{1, 2\}, \{1, 2, 3\}$
$\{1, 3\}$	$\{1, 3\}, \{1, 2, 3\}$
$\{2, 3\}$	$\{2, 3\}, \{1, 2, 3\}$
$\{1, 2, 3\}$	$\{1, 2, 3\}$

Hasse diagram of 2^X



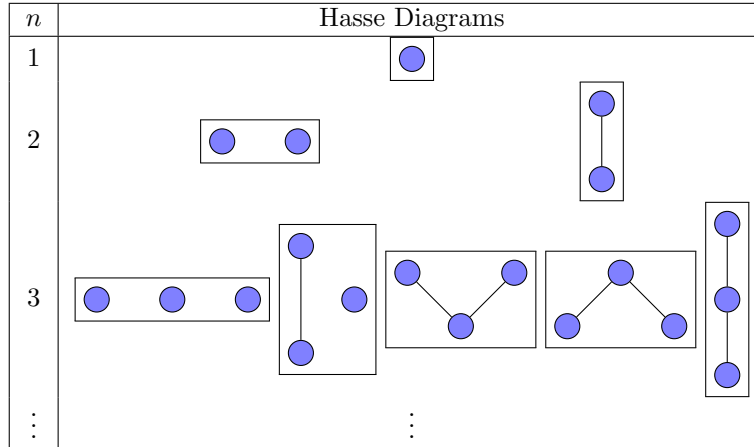
2.2 Isomorphisms of Poset

An isomorphism is represented using the symbol $\cong$ and two posets P and Q are considered isomorphic if there exists an order transferring bijection from P to Q .

Basically, $(P, \leq) \cong (Q, \leq) \iff \exists \phi: P \xrightarrow{\sim} Q. \forall x, y \in P, x \leq y \implies \phi(x) \leq \phi(y)$

A common example of isomorphic posets includes $([n], \leq) \cong ([a, a+n-1], \leq)$ the bijection is $x \xrightarrow[\phi]{\sim} a+x-1$

Note that, for all pairs of isomorphic Posets, we can trivially say that their Hasse diagrams will also be isomorphic. Using this, we can create a beautiful representation showing all the possible Isomorphism classes of Partially ordered sets, depending on the number of elements that they have.



2.3 Maximal, Minimal, Greatest and Least elements

For a Poset $(P, \leq)$ we define the following:

$x \in P$ is maximal $\iff \forall y \in P, x \leq y. y \leq x$.

$x \in P$ is minimal $\iff \forall y \in P, y \leq x. x \leq y$.

$x \in P$ is greatest $\iff \forall y \in P. y \leq x$.

$x \in P$ is least $\iff \forall y \in P. x \leq y$.

Trivially, we can say that, $\forall (P, \leq), P \neq \emptyset, n(P) < \infty. \exists \text{maximal}(P), \text{minimal}(P)$. Similarly, we can say that, there maybe at most a single greatest and least elements of a poset. Not necessary for there to be one, but if there is, then there can only be one.

2.4 Induced Subposets

Let $(P, \leq)$ and $(Q, \leq)$ be Posets s.t.

$Q \subset P$ and $\forall q_1, q_2 \in Q. q_1 \leq_Q q_2 \iff q_1 \leq_P q_2$ then Q is called an induced subposet of P .

Induced subposets are really useful for studying various isomorphism classes and their relations (how one could go to another)

For example, consider the poset $(\mathbb{P}, |)$ which is poset of Positive integers, ordered by divisibility. Subposets of $(\mathbb{P}, |)$ of the form

$p, q \in \mathbb{P}, \gcd(p, q) = 1, p \neq q. (\{p^a \cdot q^b : a, b \in \mathbb{N}\}, |)$

are isomorphic to the poset $(\mathbb{N}^2, \leq)$ where $(a, b) \leq_{\mathbb{N}^2} (c, d) \iff a \leq_{\mathbb{N}} c, b \leq_{\mathbb{N}} d$.

More generally, if we fix n distinct, pairwise coprime numbers: $(c_i)_{i=1}^n$.

Then, let

$(C_n, |)$ be $\left\{ \prod c_i^{k_i} : \forall i. k_i \in \mathbb{N}. \right\}$ and,

$(\mathbb{N}^n, \leq)$ be a poset where $\leq$ is defined by $(a_i)_{i=1}^n \leq (b_i)_{i=1}^n \iff \forall i \in [n]. a_i \leq b_i$.

Then $(C_n, |) \cong (\mathbb{N}^n, \leq)$

2.5 Incidence Algebra

Let $(P, \leq)$ be a Poset such that $\forall x, z \in P, n(y \in P : x \leq y \leq z) < \infty$

Then we define

$A(P) := \{f : P \times P \rightarrow R \mid f(x, y) \neq 0 \implies x \leq y\}$ and,

$*$: $A(P) \times A(P) \rightarrow A(P)$ as,

$$(f * g)(x, y) := \sum_{x \leq z \leq y} f(x, z) \cdot g(z, y)$$

Theorem 2.1. $*$ is associative

Proof. To prove that $\forall (P, \leq). \forall f, g, h \in A(P). (f * g) * h = f * (g * h)$.

Let's start by expanding the LHS and RHS.

LHS:

$$((f * g) * h)(x, z) = \sum_{x \leq y \leq z} (f * g)(x, y) \cdot h(y, z) = \sum_{x \leq y \leq z} \sum_{x \leq w \leq y} f(x, w) \cdot g(w, y) \cdot h(y, z)$$

RHS:

$$(f * (g * h))(x, z) = \sum_{x \leq y \leq z} f(x, y) \cdot (g * h)(y, z) = \sum_{x \leq y \leq z} \sum_{y \leq w \leq z} f(x, y) \cdot g(y, w) \cdot h(w, z)$$

Notice that

1. On the LHS, we're traversing all the quadruplets (x, y, w, z) such that $x \leq w \leq y \leq z$
2. On the RHS, we're traversing all the quadruplets (x, w, y, z) such that $x \leq y \leq w \leq z$

Which simply means that, if we swap w and y on RHS it will be equivalent to the LHS.

□

Let δ be a function in $A(P)$ such that

$$\delta(x, y) = \begin{cases} 1 & x = y \\ 0 & \text{otherwise.} \end{cases}$$

Then δ is the multiplicative unit of this algebra, i.e.

$$\forall f \in A(P). f * \delta = \delta * f = f.$$

2.5.1 Inversion of incidence algebras

Theorem 2.2. *A function f in an incidence algebra $A(P)$ of a locally finite poset P , is $*$ -invertible $\iff \forall x \in P. f(x, x) \neq 0$*

Proof. First let's start by stating the condition of theorem clearly.

$$\forall f \in A(P), (\forall x \in P. f(x, x) \neq 0) . \exists g \in A(P). f * g = g * f = \delta.$$

Where the last condition, $f * g = g * f = \delta$ can means that $\forall x, y \in P. (f * g)(x, y) = (g * f)(x, y) = \delta(x, y)$

To prove this let's look at this case by case:

Case I: $x = y$

$$\begin{aligned} (f * g)(x, x) &= \sum_{x \leq y \leq x} f(x, y) \cdot g(y, x) \\ \implies (f * g)(x, x) &= f(x, x) \cdot g(x, x) \end{aligned}$$

Which means that for g to be the inverse of f , this produce simply needs to equal 1 ($\because \delta(x, x) = 1$).

Since we assume that $f(x, x) \neq 0$ we can say that, $g(x, x) = \frac{1}{f(x, x)}$

Case II: $x < y$ For $x < y$ let's expand the function

$$(f * g)(x, y) = \sum_{x \leq z \leq y} f(x, z) \cdot g(z, y)$$

For the first value of z , note that it is $f(x, x) \cdot g(x, y)$ which is the same as $\frac{g(x, y)}{f(x, x)}$.

$\therefore$ to be an inverse $\forall x, y \in P, x < y. g(x, y) = 0$

Using the above finding we can deduce that:

$$g(x, y) = -g(x, x) \cdot \sum_{x < z \leq y} f(x, z) \cdot g(z, y)$$

$\therefore$ this is a recursive relation, which only relies on the $[x, y]$ range being computable we can use this to inductively compute this relation on any locally finite Poset.

In other words, since all the values on the RHS are defined independently of the LHS, we can say that we can go from the RHS to the LHS. $\square$

$$\boxed{\forall f, g \in A(P), f(x, x) \neq 0, g(x, y) = -g(x, x) \cdot \sum_{x < z \leq y} f(x, z) \cdot g(z, y). f * g = g * f = \delta}$$

Using the above relation we can calculate the inverse of any invertible function in $A(P)$, and it is also how we'll compute the Möbius function.